

**INDIAN INSTITUTE OF TECHNOLOGY
KHARAGPUR**

1. Name of the Academic Unit: Centre of Excellence in Artificial Intelligence

2. Subject Name: Introduction to Machine Learning

L-T-P: 3-1-0

Credits: 4

3. Pre-requisites: None

4. Syllabus and reference books:

Syllabus:

1) Pattern Recognition for Machine Learning, by Christopher M. Bishop, Springer (2006)

2) Pattern Classification, by R. Duda, P. Hart, D. Stork, Wiley (2007)

Reference Books:

1) Machine Learning, by Tom Mitchell, McGraw Hill (2017-latest edition)

2) The Hundred-Page Machine Learning Book, by Andriy Burkov (2019)

5. Lecture-wise break-up:

Sl. No.	Topic	No. of Lecture-Hours
1. Basic Concepts	Applications in different domains, basic idea of learning, feature-based representation	3 hours
2. Simple Classification Algorithms	K-Nearest Neighbor, Decision Tree	3 hours
3. Regression	Linear and regularized (ridge and LASSO) regression, logistic regression	3 hours
4. Basic Optimization	Gradient Descent, Constrained Optimization, their application to regression	3 hours
5. Linear Classifiers	Linear Separability, Perceptron, Support Vector Machine	3 hours
6. Non-linear Classifiers	Kernel Methods in SVM, multi-layer perceptron and ANN	3 hours
7. Ensemble Classification	Bagging, Random Forest, Boosting, Adaboost, Gradient Boosting	3 hours
8. Unsupervised	Clustering, hierarchical and agglomerative clustering, K-means clustering, K-means++,	5 hours

Learning	DB-scan, spectral clustering	
9. Probabilistic Approaches	Bayesian and Naïve Bayes Classifier, Gaussian Mixture Models for clustering	4 hours
10. Dimensionality Reduction	Principal Component Analysis, t-SNE, multidimensional scaling	2 hours
11. Anomaly Detection	Density-based approach, statistical approach	2 hours
12. Semi-supervised Learning	Label propagation, Laplacian-based approach	2 hours
Total number of hours		36